

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Multicentre, Event-driven, Parallel group, Phase III Study Evaluating the Efficacy and Safety of PT027 Compared with PT007 Administered As Needed in Symptomatic Chinese Adults with Asthma [BAIYUN] **Short Title/ Acronym:** BAIYUN **Protocol Number:** D6935C00003 **NCT Number:** NCT06471257 **Sponsor Name:** AstraZeneca **Investigational Product:** Budesonide/Albuterol Metered-dose Inhaler (BDA MDI / PT027 / Airsupra) **Phase of Development:** Phase III **Medical Monitor:** [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of budesonide/albuterol metered-dose inhaler (BDA MDI/PT027) compared with albuterol sulfate (AS MDI/PT007) in reducing the risk (time to first) of a severe asthma exacerbation in symptomatic Chinese adults with asthma. **Secondary Objectives:** To evaluate the severe exacerbation rate (annualised), total systemic corticosteroid exposure, and ACQ-5 responder rates.

2.2 Background & Rationale To address the risk of severe exacerbations in patients with asthma relying on short-acting beta2-agonist (SABA) alone for rescue therapy. By utilizing a fixed-dose combination of an inhaled corticosteroid and a SABA (PT027) as needed, this study aims to evaluate whether an anti-inflammatory reliever can provide a superior reduction in asthma exacerbation risk and total systemic corticosteroid use compared to standard albuterol rescue therapy in the Chinese adult population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (PT007 / Albuterol sulfate MDI) **Blinding:** Double-blind **Randomization:** Parallel group (Randomized)

3.2 Planned Enrollment & Duration Target \$N\$: 790 participants **Number of Sites:** Multicentre (China only) **Estimated Study Start:** 17-Jun-2024 **Estimated Primary Completion:** 30-Oct-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Documented physician-diagnosed asthma for at least 12 months prior to Visit 1. 3. Pre-bronchodilator FEV1 of $\geq 40\%$ to $< 90\%$ predicted normal value for adults, with documented reversibility to albuterol. 4. A documented history of at least one severe asthma exacerbation within 12 months before Visit 1. 5. ACQ-7 score ≥ 1.5 assessed at Visit 1 and ACQ-5 score ≥ 1.5 assessed at Visit 2.

4.2 Exclusion Criteria 1. Chronic obstructive pulmonary disease (COPD) or other significant lung disease. 2. Oral/systemic corticosteroid (SCS) use (any dose) within 6 weeks before Visit 1, or chronic use of OCS ≥ 3 weeks in 3 months prior to Visit 1. 3. Having received any marketed or investigational biologic within 3 months or 5 half-lives before Visit 1. 4. Current smokers, former smokers with > 10 pack-years history, or former smokers who stopped smoking < 6 months before Visit 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Arm 1: BDA MDI (Budesonide/Albuterol) 160/180 μg administered as needed (prn) for asthma symptoms or prior to exercise. Arm 2: AS MDI (Albuterol sulfate) 180 μg administered as needed. **Route of Administration:** Oral Inhalation (Metered-dose inhaler). **Duration of Treatment:** Event-driven study duration; minimum of 24 weeks and maximum of 52 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Scheduled asthma maintenance therapies (must be stable for at least 4 weeks before Visit 1). **Prohibited:** Use of spacers for MDI administration; investigational or marketed biologic therapies; systemic corticosteroids for chronic conditions (≥ 3 weeks).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -14	Informed Consent, Medical History, Asthma diagnosis confirmation, FEV1, ACQ-7 evaluation
Baseline (Run-in)	Day -14 to 0	Randomization, ACQ-5 evaluation, MDI technique assessment, Provision of rescue inhaler
Interim	Weeks 4 to 52	Efficacy & safety assessments, Exacerbation monitoring, Concomitant medication review
End of Study	Week 24 - 52	Final spirometry, Exit Interview, IP Accountability (Event-driven termination)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to first severe asthma exacerbation. **Measurement Method:** Clinical assessment of exacerbation events based on systemic corticosteroid requirements, emergency room visits, or hospitalizations over the study period.

7.2 Secondary & Exploratory Endpoints 1. Severe exacerbation rate (annualised). 2. Total systemic corticosteroid exposure (cumulative dose). 3. ACQ-5 responder rate.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient diaries, symptom logs, and investigator evaluation at scheduled visits. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, particularly asthma-related hospitalizations or life-threatening bronchospasm. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to oversee safety and trial progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analyses; Safety Set for AE/SAE tabulations. **Sample Size Justification:** Target sample of $N=790$ properly powered to detect a statistically significant difference in the time to first severe exacerbation (event-driven). **Handling of Missing Data:** Methods such as Multiple Imputation (MI) for missing continuous parameters, with time-to-event analyses appropriately censoring lost-to-follow-up patients.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to verify protocol adherence, accurate IP accountability, and proper MDI technique documentation. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing verified data clarification workflows.

11. Study Identifiers & Governance

Primary ID: D6935C00003 **Registry Number:** NCT06471257 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** BAIYUN **Official Regulatory Title:** A Randomized, Double-blind, Multicentre, Event-driven, Parallel group, Phase III Study Evaluating the Efficacy and

12. Clinical Framework (Asthma Specific)

Primary Condition: Asthma **Diagnosis Threshold:** Documented physician-diagnosed asthma ≥ 12 months, Pre-bronchodilator FEV1 of $\geq 40\%$ to $< 90\%$ predicted normal. **Medical Methodology:** Rescue/reliever therapy utilization. **Optimization Target:** Reduction in the incidence of severe exacerbations and overall systemic corticosteroid burden.

13. Study Procedures & Methodology

Management Pathway: As-needed usage of reliever medication in response to symptoms. **Education Protocol (HCP):** Site-level training on standardized exacerbation documentation and proper lung function (PEF/FEV1) assessments. **Education Protocol (Patient):** MDI administration technique verification (use of spacers strictly prohibited). **Quality Audit Mechanism:** Investigator evaluation of patient inhaler technique and accurate daily symptom/rescue use logging.

14. Observation & Follow-Up Schedule

Total Duration: 24 to 52 Weeks (Event-driven). **Onsite Visit Cadence:** Pre-defined schedule per protocol throughout the up-to-52-week treatment phase. **Remote Monitoring Frequency:** Routine data entry audits and remote checks. **Checklist Review Interval:** Regular assessment of symptom severity, rescue medication usage, and ACQ questionnaires at designated milestones.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				